REPORT OF DATA SCIENCE FOR ECONOMICS

Francesco Lassi

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1) OBJECTIVE OF THE ANALYSIS **3**

2) DATA DESCRIPTION **3**

3) PRELIMINARY ANALYSIS **3**

4) DATA MODELING **5**

a) TRAINING/ TEST SET *5*

b) MANUAL SELECTION *5*

c) STEPWISE *6*

d) REGULARIZED REGRESSION *6*

e) RPART *7*

f) BAGGING *9*

g) RANDOM FOREST *10*

h) ROC CURVES *10*

i) CUT-OFF SELECTION *11*

5) PREDICTION **12**

6) CONCLUSION **12**

1) OBJECTIVE OF THE ANALYSIS

The purpose of the report is to predict if the customer of the certain insurance companies will pay or not the premium within 90 days of due date flag. We will foresee the data contained in the “Live Data”, and to reach this aim we will analyse the “Original Data”. We will use the Original Data to arrive at a model that it is the best to predict the dependent variable as close as possible.

We will use different approaches. Our independent variable is VAR38 (dummy):
-if 0, the client doesn’t pay the premium within 90 days of due date flag
-if 1, the client pays the premium within 90 days of due date flag

I care to remember that the result is a prediction. To analyse the data, we will use R, so everything about model and programming language refers to it.

2) DATA DESCRIPTION

As we said: we have two different datasets: the Original one (to exploit) and the Live one (to predict).
The Original Data contains 28 variables and 9448 observations, the Live Data provides 27 variables (there is not the VAR38 that we want to predict) and 2929 observations.
I reported below the definitions of each variable, the type of variable (numeric, intermediate, character) and the first values of each variable, regarding Original Data.

VAR1: Masked Policy Identifier- int 20233285 20123004 20066468 20469654 20793015 19951082 20338987 20053581 20019850 20054031 ...
VAR4: Application Life Assured Alcohol Declaration- chr "N" "N" "N" "N" ...
VAR5: Applicant's Policy Annualized Premium- num 90000 25000 99999 50000 40000 ...
VAR6: Mapped Agent Vintage- int 78 33 39 39 24 16 57 32 63 36 ...
VAR7: Auto Debit of Premium Opted Flag (Yes/No)- chr "N" "N" "Y" "Y" ...
VAR9: Total Risk Exposure of HDFC Life w.r.t. Life Assured- num 1750000 250000 999990 968384 1200000 ...
VAR10: Application Sourcing Channel- chr "Ex-HDFC" "HDFC BANK" "HDFC BANK" "HDFC BANK" ...
VAR13: Application Life Assured City Tier- chr "Tier III" "Tier III" "Tier III" "Tier I" ...
VAR14: Application Life Assured Age- int 37 36 34 47 32 54 44 41 38 33 ...
VAR15: Application Life Assured Education- chr "Graduation" "B A" "MBA" "Under Matric (Class l to lX)" ...
VAR16: Application Life Assured Gender- chr "Male" "Male" "Male" "Male" ...
VAR17: Application Life Assured Income- num 400000 285000 600000 700000 600000 650000 1500000 300000 300000 400000 ...
VAR18: Application Life Assured Industry- chr "Other" "Business" "Manufacturing" "Manufacturing" ...
VAR19: Application Life Assured Marital Status- chr "Married" "Married" "Married" "Married" ...
VAR21: Application Life Assured Occupation- chr "Self-employed/ Business" "Self-employed/ Business" "Salaried" "Salaried" ...
VAR22: Applicant's Policy PAR/NON PAR/ULIP Tag- chr "UL" "UL" "UL" "Par" ...
VAR24: First Premium Payment Type- chr "DD" "Online-Bill" "Online Netbanking" "ECS,SI" ...
VAR25: Applicant's Policy Product Category- chr "Investment" "Investment" "Investment" "Savings" ...
VAR26: Applicant's Policy Premium Payment Frequency- chr "Monthly Premium" "Annual" "Annual" "Annual" ...
VAR27: Applicant's Policy Product Name- chr "HDFC SL ProGrowth - Flexi" "HDFC SL ProGrowth Super II" "HDFC Life ProGrowth Plus" "HDFC Life Super Income Plan" ...
VAR29: Applicant's Policy Price Sensitivity (1-'Yes', 0-'No')- int 0 0 0 0 0 0 0 0 1 0 ...
VAR31: Applicant's Policy Risk Cessation Term- int 10 15 10 24 10 10 15 15 17 10 ...
VAR32: Applicant's Policy Rider Opted Flag (1-'Yes', 0-'No')- int 1 0 0 1 1 0 0 1 1 ...
VAR33: Application Life Assured Smoker Declaration- chr "N" "N" "N" "N" ...
VAR34: Application Life Assured State- chr "Kerala" "Uttar Pradesh" "Tamil Nadu" "Gujarat" ...
VAR36: Applicant's Policy Sum Assured- num 1750000 500000 1999980 468384 2400000 ...
VAR37: HDFC Life Operational Zone- chr "South 2" "Unknown" "Unknown" "West" ...
VAR38: Target - Paid Premium Within 90 Days of Due Date Flag (1-'Yes', 0-'No')- int 1 1 1 1 1 1 1 1 1 0

3) PRELIMINARY ANALYSIS

This paragraph will explain rapidly some features and relationships in the Original Data.

- Dependent Variable (VAR38)

The table 1 shows the absolute frequency of the VAR38. Talking about relative frequencies, we have 79.79% of "1", and 21.20% of "0". The data are unbalanced.

var38	
0	1
2003	7445

(Table 1)

- VAR29 (Applicant's Policy Price Sensitivity (1- 'Yes', 0- 'No')- Dummy)

I report the relative frequencies of this variable (table 2) and the relationship with the dependent variable using a table (table 3).

The table 2 shows the data about the VAR29 are really unbalanced. There is a majority of "0", equal to the 94.82%.

The table 3 shows the relationship between VAR38 and VAR29 and we notice in 7031 cases VAR29=0 and VAR38=1, against the fact that in 75 cases VAR29=1 and VAR38=0.

var29	
0	1
94.824301	5.175699

(Table 2)

	VAR38	
VAR29	0	1
0	1928	7031
1	75	414

(Table 3)

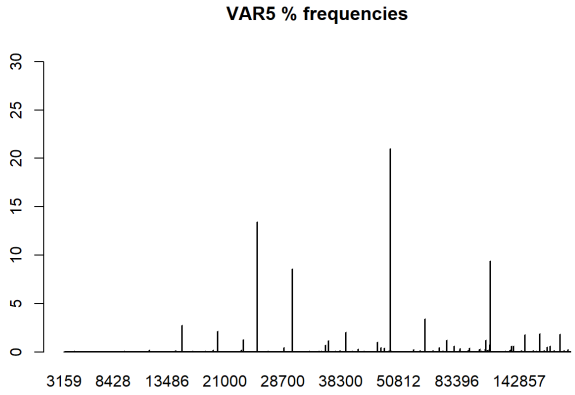
- VAR5(Applicant's Policy Annualized Premium- Numeric)

I report the most important features of this variable, the relative frequencies of the variable using a graph (figure 1), and the relationship with VAR38 (figure 2).

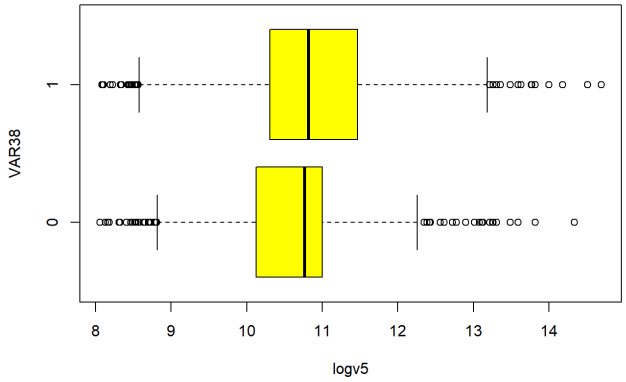
The figure 1 shows the data are balanced. Most of the data are close to the mean.

The figure 2 indicates the relationship between VAR38 and log (VAR5). It shows several outliers in the two groups. The median is almost the same. The range of the boxplot when VAR38=0 is wider and even the box, while the box when VAR38=0 is more crushed.

Min=3159 Max=2400000 Mean=73821.67 Median=50000



(figure 1)



(figure 2)

4) DATA MODELING

a) TRAINING/ TEST SET

Original Data (so with 28 variables) are divided in two different sets: Training Data and Test Data.
Test Set:1889, (20% of the Original Data)
Training Set: 7559 (80% of the Original Data)
The Training Set is used to create the fitted models. The Test Set aims to check the model performance evaluation of the fitted model. The split happened randomly and there aren't repeated observations in neither the sets.

b) MANUAL SELECTION

This section shows the results of a model built on linear regression. It analyses the opposite cases:
-Null model, the model with the only intercept
-Full model, the model with all the variables

(The results will show the statically significance of the coefficients. '***' means the coefficient is statistically significant at 0.01%, '**' mean the coefficient is statistically significant at 1%, '*' mean the coefficient is statistically significant at 5%.)

I. NULL Model

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	1.3197	0.0282	46.8	<2e-16 ***

logLik	AIC	BIC
-3893.739	7789.478	7796.408

(table 4)

The table 4 shows the estimated coefficient of the intercept and its standard error, z value and p-vale. It shows also the log-likelihood, AIC and BIC values. The coefficient of the constant is significant at 0.1%. It coincides with the estimated value of y because it is null model, but it is not the mean, given the fact VAR38 is dummy.

II. FULL Model

The linear regression in the full model that we built it is with all the variables without any transformations (as log, square, etc.). It is just composed from variables and their betas values.

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	1.111e+01	3.032e+00	3.665	0.000247 ***
VAR321	-7.178e-01	8.817e-02	-8.141	3.92e-16 ***
VAR1	-4.854e-07	1.366e-07	-3.553	0.000381 ***
VAR7Y	3.701e+00	1.098e-01	33.692	< 2e-16 ***
VAR26Halfyearly Premium	6.358e-01	2.342e-01	2.715	0.006633 **
VAR26Monthly Premium	1.109e+00	2.696e-01	4.113	3.90e-05 ***

logLik	AIC	BIC
-2527.930	5287.860	6091.798

(table 5)

The table shows only the variables that are statistically significant at 0.1% and, as usual, the values of the log-likelihood, AIC, BIC. The coefficient of the intercept is much lower than previously but still significant at the same level. Anyways, the intercept is not interpretable because there are some other coefficients in the model who cannot equal to zero. The coefficients of the VAR32 (when it is equal to 1) and the VAR1 are negative so they will decrease if VAR38 rises, the opposite is for the other variables that table 5 shows.

c) STEPWISE

Regarding Stepwise model, we have three different approaches: forward, backward and both. We will show only the last one because they give back the same results.

The only hyper-parameter is K, the penalty coefficient of the number of the variables. We chose K equal to the logarithm of the number of observations which corresponds to use BIC.

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	1.286e+01	2.101e+00	6.120	9.38e-10 ***
VAR7Y	3.580e+00	1.061e-01	33.733	< 2e-16 ***
VAR321	-6.758e-01	7.108e-02	-9.508	< 2e-16 ***
VAR1	-5.589e-07	1.028e-07	-5.435	5.47e-08 ***
VAR25Investment	-7.621e-01	1.965e-01	-3.878	0.000105 ***
VAR25Protection	4.241e-02	2.451e-01	0.173	0.862643
VAR25Savings	-8.827e-01	2.091e-01	-4.222	2.42e-05 ***
VAR13Tier II	-2.500e-01	9.952e-02	-2.512	0.012008 *
VAR13Tier III	-3.572e-01	8.204e-02	-4.354	1.33e-05 ***

logLik	AIC	BIC
-2662.861	5343.723	5406.097

(table 6)

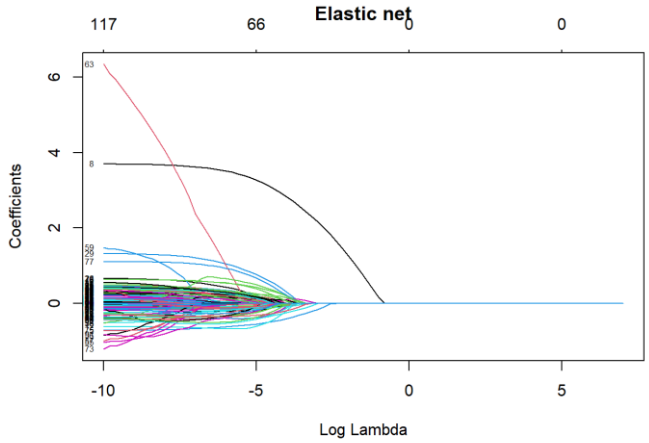
The table 6 contains the results from the Stepwise- both model and the respective values of log-likelihood, AIC and BIC. About those, we notice they are the lowest between the models which we analysed for now, except for AIC (it is lower in full model). The intercept is still significant and even VAR1, moreover the table 6 indicates the variables are significant now and with the full model they weren't.

d) REGULARIZED REGRESSION

The Regularized regression has three different approaches: Ridge, Lasso, Elastic Net. We will report only the Elastic Net because it is a synthesis of the other two applications. The Regularized Regression doesn't consider the useless parameters, their beta will equal to zero.
The aim of this application is to minimize the following equation:

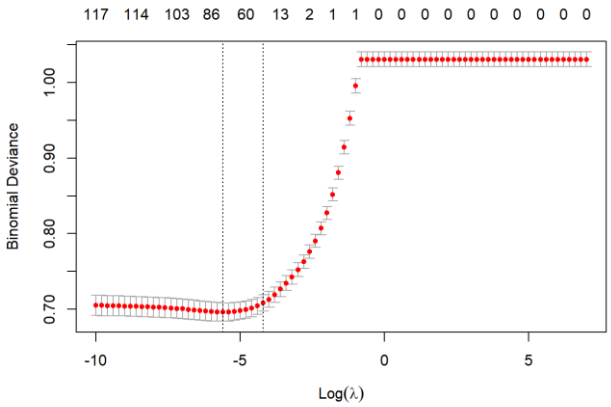
$$\beta\,(\alpha,\,\lambda) = min\sum_{i=1}^{N}[y_i\,lnf(x_i) + (1-y_i)\,lnf(x_i) + \lambda\,[(1-\alpha)\,\sum_{j=1}^{p}\beta j\,2 + \alpha\sum_{j=1}^{p}|\beta_j\,|]$$

The first step is to choose the alpha (the hyper-parameter) that we want to use based on the type of Regularized Regression. So, for Elastic Net, we chose alpha equal to 0.5.



(figure 3)

The Figure 3 shows the values coefficients based on the varying the log (λ). It indicates higher is the lambda value, higher is its log (λ), the betas of the coefficients tend to go to zero. We have to select the best lambda through Cross Validation.



(figure 4)

The Figure 4 shows from a graphical point of view the selection of the best lambda. The best lambda is the value on the abscissa that corresponds to the second dash line from right to left. After this last process, the model has been refitted with the value of this best lambda.

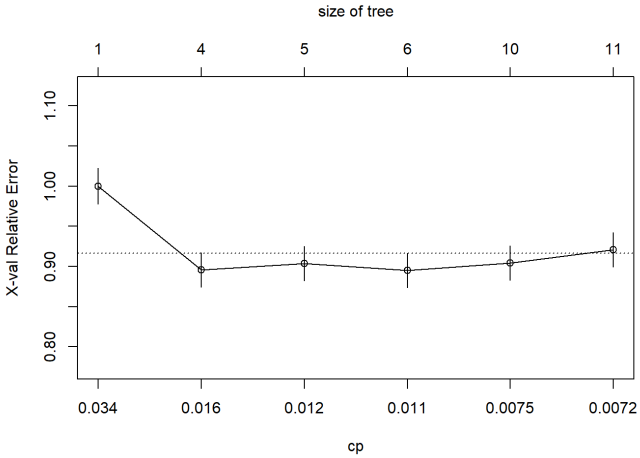
e) RPART

Rpart is an algorithmic approach to study the data. It gives back as output a tree. In this approach, we have to define some hyper-parameters to reach our aim.

	CP	nsplit	rel error	xerror	xstd
1	0.0338770	0	1.00000	1.00000	0.022250
2	0.0163112	3	0.86700	0.89586	0.021351
3	0.0119197	4	0.85069	0.90339	0.021419
4	0.0112923	5	0.83877	0.89523	0.021345

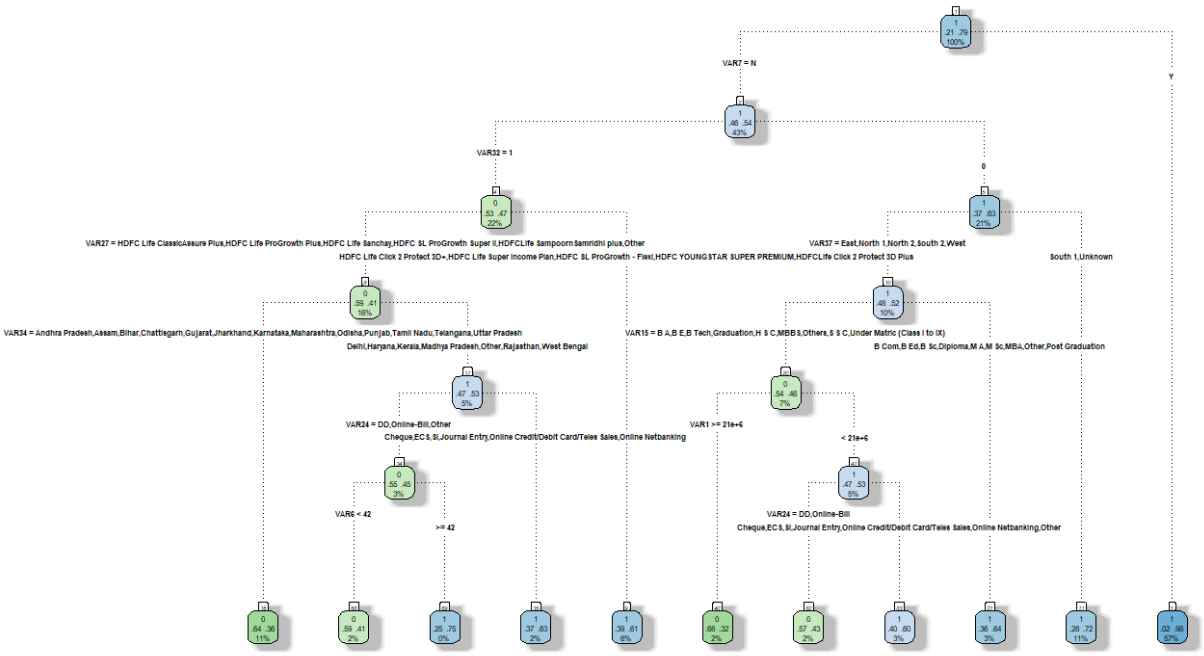
5 0.0075282 9 0.79046 0.90402 0.021425
6 0.0072146 10 0.78294 0.92095 0.021577 (table 7)

The table 7 contains the value of the CP, the number of splits, the relative errors, the xerrors and the standard deviation for each step. The relative errors decrease for each step. There is the opposite situation for the xerrors and even for the standard deviation.



(figure 5)

In the figure 5, the selection of the best CP value has showed. It corresponds to the intersection between the horizontal dashed line and the sloped line. The selection process of the CP value is the same explained above about lambda selection. To build the tree, this value will be used. In the table 7, the CP value decrease for each step.



(figure 6)

The variables actually used in tree construction: VAR1, VAR15, VAR24, VAR27, VAR32, VAR34, VAR37, VAR6, VAR7. In each leaf the tree shows observations that have specific features defined by the development of the tree. Every leaf is built with an intersection of variables intervals. For example, looking at the first split, the observations are divided in

observations which have VAR7=N (they are 47% of the total observations), and in observations which have VAR7=Y (they are 53% of the total observations).

f) BAGGING

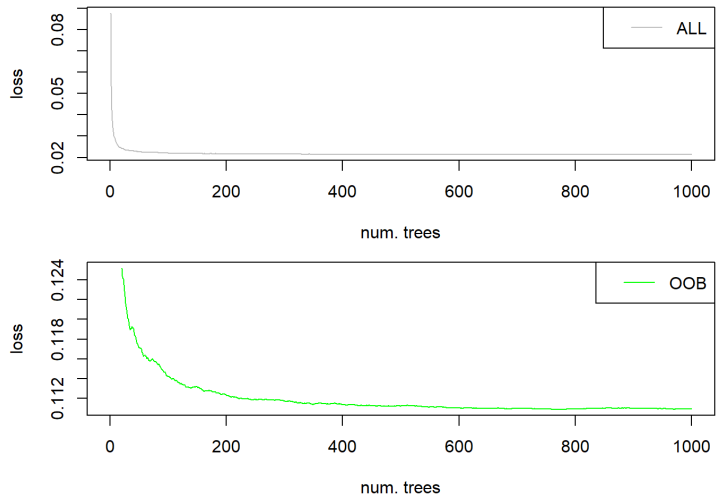
Bagging is an approach that applies the Rpart to small samples. This approach is better than Rpart, because it reduces prediction variance, so improving predictions performance on live data.

For Bagging, we have to set some hyper-parameters:
- number of trees specifies the number of samples. Higher values of it are better, but not too much otherwise the prediction accuracy decreases. For the analysis is put equal to 1000.
- mtry: the number of features randomly selected before a split (for Bagging is always put equal to the number of all the features eligible for the split).
-min.bucket=1
-min.node.size=5
- max.depth: NULL (split without any limit)
The last three hyper-parameters are settled using the default choices.

	me	ce	brier
ib	0.00000000	0.02438423	0.001545022
oob	0.17581691	0.34089791	0.110954241
all	0.00251356	0.10899532	0.021351240

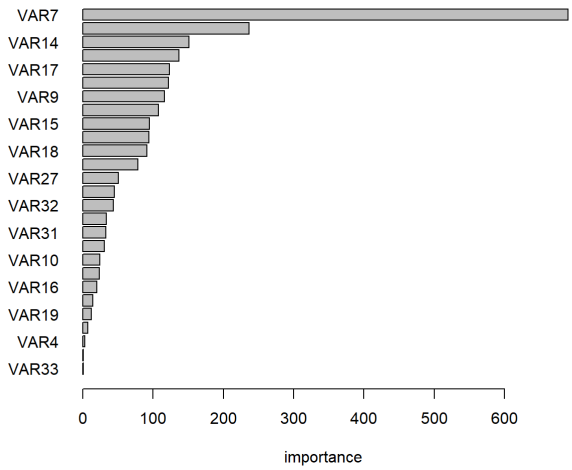
(table 8)

The table 8 shows the miss-classification error (ME), the brier, cross-entropy (CE). They are all error measures for ib, oob, and for all the observations.
ME is equal to 0 in ib because it's impossible to fail predictions on data that we already have. These error measures are important to try to find the best number tree's value. The best number tree's value is the one which gives back the same error measure's value if we try to rise the number of the samples.



(figure 7)

The figure 7 shows the trend of brier by varying the number of trees. Loss means "brier". In the figure 7, in the first plot, the best number of trees is 200, on the other plot, for oob, the best number of trees is 1000 (as we set in parameter at the beginning).



(figure 8)

In the figure 8, we can see how much every variable is important. The most important one is VAR7, the second one is VAR14, and the third one is VAR17, after this the importance is going down slowly. The other models, as manual full model and stepwise model are considering VAR14 and VAR17 not so relevant.

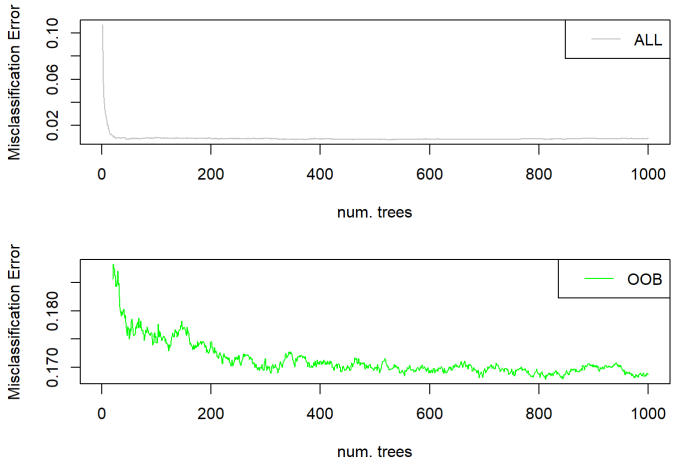
g) RANDOM FOREST

The difference with Bagging is Random Forest uses a different mtry equal to the square of features number. Because of this it works differently.

	me	ce	brier
ib	0.000000000	0.04678092	0.00446342
oob	0.168805398	0.34225921	0.11007962
all	0.008731314	0.12957354	0.02670950

(table 9)

The table 9 shows the same measure errors of the table 8, but these values are regarding obviously Random Forest.

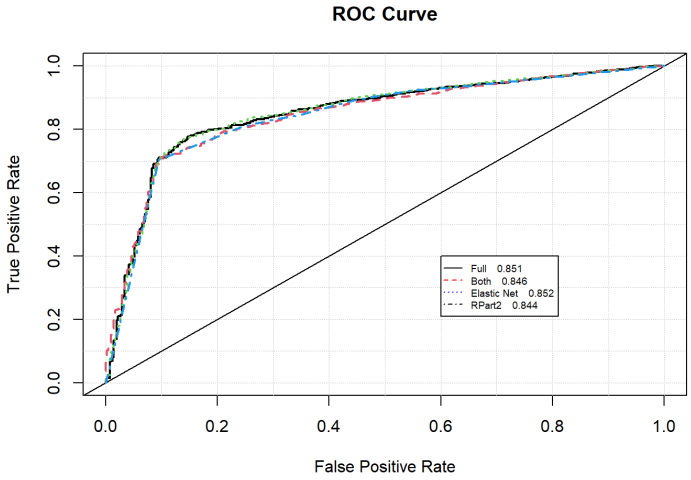


(figure 9)

The figure 9 shows how the Misclassification error is changing based on the varying of the number of trees. In all (the first plot), the number of trees is still 200, but in the second plot, oob, it is not possible to determine the number of trees. We don't show the graph about the importance of each variable because he has the same result of Bagging approach.

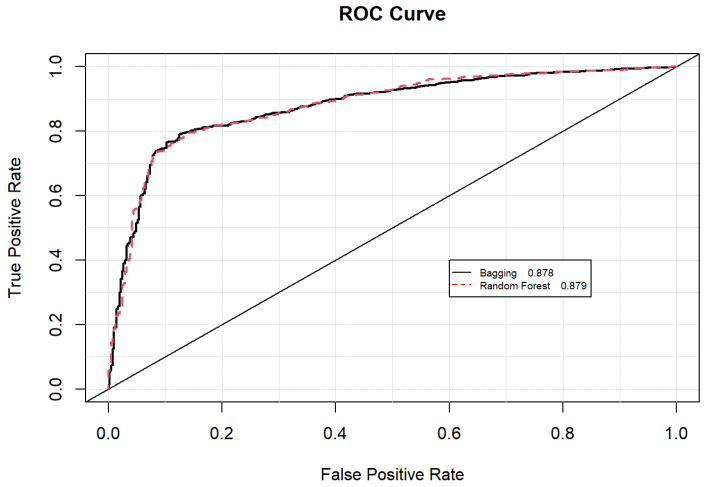
h) ROC CURVES

ROC Curve is an application to show the model performance evaluation. ROC Curve exploits the relationship among test TPR (true positive rate) and FPR (false positive rate) by the varying of the cut-off value. When we are analysing a binary dependent and we are applying it on fitted models to predict the same variable, from the prediction we should obtain the probability to estimate the dependent variable equal to 1. To obtain this result, we have to transform the final results in either 0 or 1 output with the "classification". The cut-off value is a value which we have to set and it aims to which predictions will be 0 or 1: if the probability prediction is major than cut off value, estimates will be 1, otherwise 0.



(figure 10)

The Figure 10 shows the ROC curve computed for the full model, for Stepwise-both, for Elastic Net and for Rpart, with the best lambda. The are under the ROC curve is called AUROC and the legend in the plot shows these values. Higher is the AUROC, better is the prediction. In the figure 10, we have the best situation with the Elastic Net.



(figure 11)

The Figure 11 shows the ROC curve for Bagging and Random Forest. The legend shows the AUROC regarding Bagging and Random Forest. The one for Random Forest is the greatest one between the model, so with the Random Forest we should have the best predictions. It is obvious that Random Forest has better value of Bagging, because it is an improvement of it. So, for our predictions we should use the Bagging, because it is the best model by AUROC point of view and even because the difference in AUROC with Random Forest is not so important.

i) CUT-OFF SELECTION

	0	1
data.train	0.21087445	0.7891255
class.av	0.29448340	0.7055166
class.qn	1.00000000	1.0000000
class.cost	0.06323588	0.9367641
class.yuden	0.21087445	0.7891255

(table 10)

The table 10 shows the 0/1 portion in data train for each cut-off value's type.
The cut off value that we selected to make the predictions is the Yuden point and his value is 0.5362.

	true	
class	0	1
0	259	168
1	150	1312

(table 11)

The table 11 shows the confusion matrix. It indicates the absolute frequencies of the predictions on Test Set using Bagging model with the selected cut off value. The columns are the true outcomes values, and the rows are the classification. The predictions are pretty good, when the true value is 1 (the right ones are 1312 and the wrong ones are 150). When the true is equal to zero the right ones are 259 and the wrong ones are 168.

5) PREDICTION

To make the predictions, as we said, we used the Bagging model with the cut off value equal to Yuden point. The results of predictions are in the file “Data-livePredicitions.txt”. We used Bagging, because, as we said, it has the best AUROC.

6) CONCLUSION

To conclude, in this document, we used, and we analysed different approaches and application. We started to take a general look of the data, going deeper and trying to reach our purpose: predict the timely payment. We discussed the most important points regarding each model which we presented, and we exploited the ROC curve to check which model was the proper one for our aim. We chose Bagging and based on confusion matrix; it gives back a good prediction based on the data that we have.